ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 4: The Incommensurable Variable

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did toy algebra tiles from 1990 make x = 3, and why did that ruin student understanding?'
- Ask: 'Can you ever combine 2x + 3 into 5x? Why do the physical pieces prevent that?'
- Ask: 'What does the area of a tile represent algebraically?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to force 3.5 units to equal x. Explain that x is a continuous variable that can represent any real number.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"If x was 10, how many unit tiles would balance one x-rod in weight? If x was 100?"
""

OFFICIAL SOLUTION KEY:

If x=10, 10 units. If x=100, 100 units. The physical rod represents the variable independent of its assignment.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

This Table Tent is certified for perpetual photocopying, laminating, and distribution within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 1: The Incommensurable Variable

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why can't you just count unit squares to find the value of x? In this lab, you prove that x is physically incapable of being an integer.

STEP-BY-STEP APPARATUS SETUP:

1. Pull one x-rod (51.0 × 14.0 mm) and several unit tiles (14.0 × 14.0 mm) from your kit.
2. Try to line up unit tiles along the length of the x-rod.
3. Notice: Three unit tiles (42.0 mm) are too short; four unit tiles (56.0 mm) are too long!

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED:	Algebra Tiles: 1 x^2 flat (51×51mm), 1 x-rod (51×14mm), unit tiles (14×14mm)
DATUM TARGET:	x = 51.0 mm, Unit = 14.0 mm -> Ratio = 3.6428... != Integer (Kills x=2 Fallacy)
CALCULATOR KEY:	[x] [ENTER] [3] [+] -> x + 3

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 4 (Algebraic Expressions)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Variables & Patterns)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 7, Ch. 3 • CCSS.MATH.7.EE.A.1 • TEKS 7.10A
GRADE LEVEL BAND:	Grades 7-9 • Target Time: 15 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 3 ('Algebra & Transcendentals'), pp. 48-62
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 2 (Incommensurability Proof) & Page 5 (Setup)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 2: LAB INVESTIGATION: THE INCOMMENSURABILITY PROOF

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Measure: 3 units = 42.0 mm. 4 units = 56.0 mm. The x-rod is exactly 51.0 mm.

Step 2: Conclude: x is NOT equal to 3, and x is NOT equal to 4. It cannot be counted in units!

Step 3: Now compare the width of the x-rod to a unit tile: both are exactly 14.0 mm.

Step 4: This means an x-rod has dimensions 1 unit wide by x units long, giving it an AREA of exactly 1 × x = x.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

- ✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
- ✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
- ✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
- ✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
x = 51.0 mm, Unit = 14.0 mm –> Ratio = 3.6428... != Integer (Kills x=2 Fallacy)

PHYSICAL APPARATUS MANIFEST:

Algebra Tiles: 1 x^2 flat (51×51mm), 1 x-rod (51×14mm), unit tiles (14×14mm)

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 1/6 | FOUNDATIONAL | 15 Min

PANEL 3: LEVEL 3 CHALLENGE: THE x^2 FLAT VERIFICATION

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE x^2 FLAT VERIFICATION

THE NON-ROUTINE PUZZLE:

Place the large square tile (x^2 flat) on your desk. Prove that its area is exactly x × x.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Place an x-rod along the horizontal edge of the x^2 flat.**
   Expected Physical Finding: **Length matches exactly (51.0 mm)**
2. **Place a second x-rod along the vertical edge of the x^2 flat.**
   Expected Physical Finding: **Height matches exactly (51.0 mm)**
3. **What is the algebraic area of this tile?**
   Expected Physical Finding: **Area = x · x = x^2**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [ENTER] [3] [+] –> x + 3

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x	X: x	Variable operand x
ENTER	Y: x | X: x	Stack lift
3	Y: x | X: 3	Constant operand 3
+	X: x + 3	Expression evaluated symbolically

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 4: The Distributive Area Model

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why does multiplying x by x produce a square tile?'
- Ask: 'How does this area model prove that x(a + b) = xa + xb?'
- Ask: 'What would happen if you tried to fill the corner with unit tiles instead of an x^2 flat?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students adding lengths to areas (e.g., claiming the area is x + x + 3). Remind them: boundary rails show LENG TH; interior tiles show AREA.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"A rectangle has an area of $3x^2 + 6x$. If its width is 3x, what is its physical length?"
""

OFFICIAL SOLUTION KEY:

Length = $(3x^2 + 6x) / 3x = x + 2$.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 1: The Distributive Area Model

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Multiplying algebra terms is not just shuffling letters on a page—it is building the surface area of a physical rectangle.

STEP-BY-STEP APPARATUS SETUP:

1. Snap the two L-frame coordinate rails together at a 90° right angle on your desk.
2. Along the vertical axis, lay down one x-rod (length x).
3. Along the horizontal axis, lay down one x-rod and three unit tiles (length x + 3).

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: L-Frame Coordinate Rails, 1 x^2 flat, 5 x-rods, 6 unit tiles

DATUM TARGET: Area Rectangle: Width = x, Height = (x + 3) -> Total Area = $x^2 + 3x$

CALCULATOR KEY: [x] [ENTER] [x] [3] [+] [*] -> x^2 + 3x

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 7A, Chapter 5 (Algebraic Expansion)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6, Module 2 (Area Model for Multiplication)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Grade 8, Ch. 7 • CCSS.MATH.HSA.APR.A.1 • TEKS A.10D
GRADE LEVEL BAND:	Grades 7–9 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 3, pp. 62–74
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 5 (L-Frame Coordinate Setup) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 2: LAB INVESTIGATION: FILLING THE COORDINATE STAGE

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Your axes define a rectangle with dimensions x by (x + 3).

Step 2: Fill the interior corner: Place one x^2 flat at the corner (dimensions x by x).

Step 3: Fill the remaining space to the right: Place three x-rods side-by-side (each 1 by x).

Step 4: Count the physical pieces inside the rectangle: Exactly one x^2 tile and three x-rods! Total area = $x^2 + 3x$.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Area Rectangle: Width = x, Height = (x + 3) -> Total Area = $x^2 + 3x$

PHYSICAL APPARATUS MANIFEST:

L-Frame Coordinate Rails, 1 x^2 flat, 5 x-rods, 6 unit tiles

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 2/6 | FOUNDATIONAL | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: EXPANDING 2x(x + 2)

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: EXPANDING 2x(x + 2)

THE NON-ROUTINE PUZZLE:

Set the horizontal axis to (x + 2) and the vertical axis to 2x (two x-rods). Fill the rectangle.

FORM A MASTER STATION TASKS (Complete All 3):

1 **How many x^2 flats fit inside the frame?**
Expected Physical Finding: **Two x^2 flats ($2x^2$)**

2 **How many x-rods fit beside them?**
Expected Physical Finding: **Four x-rods (4x)**

3 **Write the expanded algebraic identity.**
Expected Physical Finding: **$2x(x + 2) = 2x^2 + 4x$**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [ENTER] [x] [3] [+] [*] -> x^2 + 3x

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x	X: x	Width operand
ENTER	Y: x | X: x	Stack lift
x+3	Y: x | X: x + 3	Height operand
*	X: $x^2 + 3x$	Multiply polynomials via distribution

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 4: Binomial Multiplication

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why can the 3 horizontal rods and 2 vertical rods be added together into 5x?'
- Ask: 'Why can\'t the 6 unit tiles be added to the 5x rods?'
- Ask: 'How does this physical model make sign errors impossible?'

⚠️ COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):

Watch for students thinking (x + 2)(x + 3) equals x² + 6 (forgetting the middle term). Point directly to the 5 x-rods oc
cupying the middle zones.

★ FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):

"Model (x + 4)(x + 1) using tiles. What are the 4 quadrant pieces, and what is the final polynomial?"
""

OFFICIAL SOLUTION KEY:

Quadrants: x², 1x, 4x, 4 units -> Simplified: x² + 5x + 4.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

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step-by-step video solution, and printable rubric.

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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 1: Binomial Multiplication

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why does FOIL (First, Outside, Inside, Last) work? Because every binomial multiplication forms four physical quadrant zones in a rectangle.

STEP-BY-STEP APPARATUS SETUP:

1. Set the horizontal axis of the L-frame to (x + 3): one x-rod + three unit tiles.
2. Set the vertical axis of the L-frame to (x + 2): one x-rod + two unit tiles.
3. Prepare to fill the entire rectangular space enclosed by the rails.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: L-Frame Rails, 1 x² flat, 5 x-rods, 6 unit tiles

DATUM TARGET: Area: (x + 2)(x + 3) = x² + 3x + 2x + 6 = x² + 5x + 6

CALCULATOR KEY: [x] [2] [+] [ENTER] [x] [3] [+] [*] -> x^2 + 5x + 6

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH: Singapore Dimensions: Grade 8A, Chapter 3 (Expansion of Binomials)

BRIDGES IN MATHEMATICS: Bridges in Mathematics: Grade 5, Unit 6, Module 3 (Binomial Arrays)

BIG IDEAS / CCSS / TEKS: Big Ideas Math: Algebra 1, Ch. 7, Sec. 7.2 • CCSS.MATH.HSA.APR.A.1 • TEKS A.10B

GRADE LEVEL BAND: Grades 8–9 • Target Time: 20 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION: Vol. 2: SC-32 Scientific Handbook, Ch. 7 ('Polynomials & Brent Solver'), pp. 88–96

HARDWARE UNBOXING BOOKLET: ALGB-01 Booklet: Page 6 (Aha! Discovery: FOIL Area Model)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 2: LAB INVESTIGATION: THE 4-ZONE CORNER GRID

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1 (First): Fill the top-left corner where x meets x: Place one x^2 flat.

Step 2 (Outside): Fill the top-right zone where x meets 3 units: Place three horizontal x-rods.

Step 3 (Inside): Fill the bottom-left zone where 2 units meet x: Place two vertical x-rods.

Step 4 (Last): Fill the bottom-right corner where 2 units meet 3 units: Place a 2×3 array of unit tiles (6 units).

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Area: $(x + 2)(x + 3) = x^2 + 3x + 2x + 6 = x^2 + 5x + 6$

PHYSICAL APPARATUS MANIFEST:

L-Frame Rails, 1 x^2 flat, 5 x-rods, 6 unit tiles

💡 LABORATORY PRO-TIP:

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 3/6 | INTERMEDIATE | 20 Min

PANEL 3: LEVEL 3 CHALLENGE: COMBINING LIKE PIECES

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: COMBINING LIKE PIECES

THE NON-ROUTINE PUZZLE:

Gather and group all the identical physical tiles inside the completed rectangle.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Count the x^2 flats.**
   Expected Physical Finding: 1 (x^2)
2. **Combine all the x-rods: 3 horizontal + 2 vertical.**
   Expected Physical Finding: 5 x-rods (5x)
3. **Count the unit tiles in the corner.**
   Expected Physical Finding: 6 unit tiles (+6)
4. **Write the final simplified polynomial.**
   Expected Physical Finding: $x^2 + 5x + 6$

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [2] [+] [ENTER] [x] [3] [+] [*] -> x^2 + 5x + 6

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x+2	X: x + 2	First binomial factor
ENTER	Y: x + 2 | X: x + 2	Stack lift
x+3	Y: x + 2 | X: x + 3	Second binomial factor
*	X: $x^2 + 5x + 6$	Polynomial multiplication

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 4: Factoring Quadratics

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did finding the factors depend on finding two numbers that add to 7 and multiply to 12?'
- Ask: 'What happens if you have a prime polynomial like $x^2 + 5x + 7$? Can it form a rectangle?'
- Ask: 'How does the physical puzzle eliminate guessing on tests?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students trying to place unit tiles directly adjacent to the x^2 flat. Remind them: only edges of matching leng
th can touch!

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Factor $x^2 + 8x + 15$ using tiles. How must the 8 rods be split to accommodate the 15 unit tiles?"
""

OFFICIAL SOLUTION KEY:

Split 8 into 3 and 5, since 3 × 5 = 15. Factors: $(x + 3)(x + 5)$.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

- National Standards: Common Core State Standards (CCSS-M)
- State Frameworks: Texas Essential Knowledge & Skills (TEKS)
- Florida B.E.S.T. & Singapore Math Institute Calibrated
- CPA Sequence: Concrete -> Pictorial -> Abstract Transfer

CLASSROOM STATION PROTOCOL & PERPETUAL REPRODUCTION RIGHTS

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within the purchasing teacher's classroom or homeschool household.
Replacement Parts & Manipulative Add-Ons: Visit stackcalc.io/parts ($2.49 Dispatch Guarantee).
Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 1: Factoring Quadratics

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Factoring is simply playing the area game in reverse: given a pile of loose tiles, can you assemble them into one perfect seamless rectangle?

STEP-BY-STEP APPARATUS SETUP:

1. Take 1 x^2 flat, 7 x-rods, and 12 unit tiles from your tray.
2. Your challenge: Arrange ALL 20 pieces into a solid, gap-free rectangle.
3. Rule: The x^2 flat must stay in the top-left corner; unit tiles must stay in the bottom-right.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: Algebra Tiles: 1 x^2 flat, 7 x-rods, 12 unit tiles + L-Frame Rails

DATUM TARGET: Reconstruct Rectangle: Area $x^2 + 7x + 12$ -> Dimensions $(x + 3) \times (x + 4)$

CALCULATOR KEY: [x^2 + 7x + 12] [FACTOR] -> $(x + 3)(x + 4)$

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8A, Chapter 3, Lesson 3.3 (Factoring Quadratics)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Factoring Polynomial Arrays)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 7, Sec. 7.5 • CCSS.MATH.HSA.SSE.A.2 • TEKS A.10E
GRADE LEVEL BAND:	Grades 8–10 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7, pp. 96–104
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 6 (Factoring Reverse Puzzle)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 2: LAB INVESTIGATION: THE RECTANGLE RECONSTRUCTION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Place the x^2 flat at the corner.

Step 2: Try splitting the 7 x-rods: Place 6 along the top and 1 along the side. Can 12 unit tiles fill a 1×6 space? (No: 1×6=6, 6 units left over).

Step 3: Try splitting 7 into 5 and 2. Can 12 units fill a 2×5 space? (No: 2×5=10, 2 units left over).

Step 4: Try splitting 7 into 3 and 4! Notice: 3 × 4 = 12! The 12 unit tiles fit into a perfect 3×4 array with zero gaps!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Reconstruct Rectangle: Area $x^2 + 7x + 12$ -> Dimensions $(x + 3) \times (x + 4)$

PHYSICAL APPARATUS MANIFEST:

Algebra Tiles: 1 x^2 flat, 7 x-rods, 12 unit tiles + L-Frame Rails

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 4/6 | INTERMEDIATE | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: READING THE DIMENSIONS

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: READING THE DIMENSIONS

THE NON-ROUTINE PUZZLE:

Lay the L-frame rails along the edges of your completed rectangle to read the factors.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Measure the horizontal edge of the rectangle.**
   Expected Physical Finding: **1 x-rod + 4 units = (x + 4)**

2. **Measure the vertical edge of the rectangle.**
   Expected Physical Finding: **1 x-rod + 3 units = (x + 3)**

3. **State the factored form of $x^2 + 7x + 12$.**
   Expected Physical Finding: **(x + 3)(x + 4)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x^2 + 7x + 12] [FACTOR] -> (x + 3)(x + 4)

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
poly	X: $x^2 + 7x + 12$	Input quadratic polynomial
FACTOR	X: $(x + 3)(x + 4)$	Root decomposition into linear binomial factors

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

CALIFORNIA LAB DESIGN // COPYRIGHT STACKCALC

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 4: Zero-Pair Annihilation

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did the x-rods completely disappear from the final expression?'
- Ask: 'Why is this identity called the DIFFERENCE of squares?'
- Ask: 'Can you ever get a sum of squares like $x^2 + 4$ using real binomials?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students thinking (−2) × (−2) equals −4. Point out that here we multiply (+2) by (−2), which produces negative tiles.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Factor $4x^2 - 9$. What are the binomial dimensions, and how many zero pairs are annihilated?"
""

OFFICIAL SOLUTION KEY:

Dimensions: (2x + 3)(2x − 3). Six (+x) rods and six (−x) rods annihilate (12 zero pairs).

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera at the QR code to launch the interactive 3D model, step-by-step video solution, and printable rubric.

https://stackcalc.io/products/algebra-boxes/#product-videos

INSTITUTIONAL CURRICULUM CERTIFICATION

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Hardware Standard: 19.60 mm pitch • Gridfinity 42.0 mm multi-cell chassis.

---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 1: Zero-Pair Annihilation

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

How can you multiply two binomials and end up with NO middle x-terms at all? In this lab, positive and negative tiles annihilate like matter and antimatter.

STEP-BY-STEP APPARATUS SETUP:

1. Pull positive tiles (mirror-ironed smooth) and negative tiles (micro-grooved ribbed).
2. Rule of Annihilation: When one positive (+1) and one negative (−1) touch, they form a 'Zero Pair' and vanish.
3. Set L-frame rails to (x + 2) vertically and (x − 2) horizontally.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: 1 x^2 flat, 2 positive x-rods (smooth), 2 negative x-rods (ribbed), 4 negative units

DATUM TARGET: Annihilation Invariant: (+x) + (−x) = 0 -> Net Area = $x^2 - 4$

CALCULATOR KEY: [x] [2] [+] [ENTER] [x] [2] [−] [*] -> x^2 − 4

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8A, Chapter 3, Lesson 3.4 (Difference of Squares)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Zero Pairs & Integer Operations)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 7, Sec. 7.7 • CCSS.MATH.HSA.SSE.A.2 • TEKS A.10F
GRADE LEVEL BAND:	Grades 8–10 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7, pp. 104–112
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 2 (Zero-Pair Tactile Texture) & Page 6 (Aha! Discovery)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: THE ZERO-PAIR EXTINCTION

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: Top-left: Place one positive x^2 flat ($+x^2$).

Step 2: Top-right: x meets (−2), producing two negative x-rods (−2x, ribbed).

Step 3: Bottom-left: (+2) meets x, producing two positive x-rods (+2x, smooth).

Step 4: Pair each positive x-rod with a negative x-rod. Notice: (+2x) + (−2x) = 0! Remove them from the board!

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Annihilation Invariant: (+x) + (−x) = 0 -> Net Area = $x^2 - 4$

PHYSICAL APPARATUS MANIFEST:

1 x^2 flat, 2 positive x-rods (smooth), 2 negative x-rods (ribbed), 4 negative units

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 5/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: THE REMAINING CORNER

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: THE REMAINING CORNER

THE NON-ROUTINE PUZZLE:

Inspect the bottom-right corner where (+2) meets (−2).

FORM A MASTER STATION TASKS (Complete All 3):

1. **What tiles fill the corner? (+2) × (−2).**
   Expected Physical Finding: **Four negative unit tiles (−4, ribbed)**

2. **What tiles remain on the entire board after zero-pairs vanish?**
   Expected Physical Finding: **One positive x^2 flat and four negative units**

3. **State the algebraic result of (x + 2)(x − 2).**
   Expected Physical Finding: **$x^2 - 4$ (Difference of Squares!)**

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x] [2] [+] [ENTER] [x] [2] [−] [*] -> x^2 − 4

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
x+2	X: x + 2	Binomial sum
ENTER	Y: x + 2 | X: x + 2	Stack lift
x−2	Y: x + 2 | X: x − 2	Binomial difference
*	X: $x^2 - 4$	Difference of squares: cross terms annihilate

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 4: Completing the Square

FACILITATOR SOCRATIC GUIDE • EXTENSIONS • ANSWER KEY

STACKCALC LEARNING LAB

TEACHER & FACILITATOR SOCRATIC GUIDE • FORMATIVE ASSESSMENT

SOCRATIC PROMPTS (Ask Your Table Team):

- Ask: 'Why did we have to divide the 6 rods into two equal halves of 3?'
- Ask: 'How does $(b/2)^2$ physically appear in the corner of the square?'
- Ask: 'How does this physical exercise turn into the quadratic formula $-b \pm \sqrt{b^2-4ac}/2a$?'

⚠️ **COMMON STUDENT MISCONCEPTION TO CATCH (DIAGNOSTIC TIP):**

Watch for students forgetting to add 9 to BOTH sides of the equation. Emphasize that adding physical tiles to the left r
equires balancing the right.

★ **FORM B EXTENSION RIDDLE (For Fast Finishers & Homework):**

"Complete the square for $x^2 + 10x$. How many unit tiles must be added to fill the corner hole?"
""

OFFICIAL SOLUTION KEY:

Split 10 into 5 and 5. Corner hole is $5 \times 5 = 25$ unit tiles.

3D VIDEO DEMO & DIGITAL SOLUTION WALKTHROUGH

SCAN FOR 3D SOLUTION

INSTANT CLASSROOM ACCESS

Point any phone, iPad, or Chromebook camera
at the QR code to launch the interactive 3D model,
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---- FOLD HERE TO CREATE SELF-STANDING TABLE TENT (STAND ON DESK) ----

ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 1: Completing the Square

STATION SETUP • TEAM ROLES • CURRICULUM SYNCHRONIZATION

STACKCALC LEARNING LAB

STATION CHALLENGE HOOK & LAB TABLE SETUP

THE MISSION:

Why is it called 'Completing the Square'? Because there is a literal, physical hole in the corner of your geometric square waiting to be filled.

STEP-BY-STEP APPARATUS SETUP:

1. Take 1 x^2 flat and 6 x-rods.
2. Your goal: Build a PERFECT geometric square.
3. Start by placing the x^2 flat at the corner.

REQUIRED HARDWARE & CALIBRATION DATUM

PARTS NEEDED: 1 x^2 flat, 6 x-rods, 9 unit tiles + SC-32 Hardware Brent Solver

DATUM TARGET: Square Completion: $x^2 + 6x + 9 = (x + 3)^2$

CALCULATOR KEY: [x^2 + 6x - 16 = 0] [SOLVE] -> x = 2, x = -8

COLLABORATIVE TABLE GROUP ROLES (Rotate Roles Each Station):

🛠️ BUILDER: Assembles the physical blocks, docks into trays, and checks level summits.
📏 MEASURER: Verifies Mitutoyo datum height measurements and reads coordinate ticks.
⌨️ CALCULATOR OPERATOR: Keys keystrokes into SC-6/SC-32 and traces register stack flow.
📝 SCRIBE / REPORTER: Records group data, verifies answers, and presents findings to class.

TRI-DIRECTIONAL CURRICULUM & BOOK SYNCHRONIZATION

1. OFFICIAL STATE CURRICULUM SYLLABI ("THE OTHER BOOKS")

SINGAPORE DIMENSIONS MATH:	Singapore Dimensions: Grade 8B, Chapter 2 (Completing the Square)
BRIDGES IN MATHEMATICS:	Bridges in Mathematics: Grade 5, Unit 6 (Geometric Square Completion)
BIG IDEAS / CCSS / TEKS:	Big Ideas Math: Algebra 1, Ch. 9, Sec. 9.4 • CCSS.MATH.HSA.REI.B.4.B • TEKS A.8A
GRADE LEVEL BAND:	Grades 9–11 • Target Time: 25 Min

2. OPTIONAL SUPPLEMENTAL STUDY (OUR REFERENCE BOOKS)

RECOMMENDED COMPANION:	Vol. 2: SC-32 Scientific Handbook, Ch. 7, pp. 112–124
HARDWARE UNBOXING BOOKLET:	ALGB-01 Booklet: Page 7 (RPN Solver Connection) & Page 8 (Exercises)

Note: Our reference manuals are optional literary and computational enrichment.
They provide historical narratives, proofs, and deep-dive problem banks.

CLASSROOM LAB STATION RULES:

1. Handle with Care: All parts are precision lab instruments calibrated to ±0.05 mm.
2. Check Level Summits: Use a straightedge or eye-level inspection for physical proofs.
3. Self-Auditing Storage: Return all blocks to labeled sockets before rotating stations.
4. Stack First, Operate Second: In RPN, place numbers into the stack before executing.
5. Turn the Tent Over: When your team finishes, flip to the inside for Level 1, 2, and 3 tasks!

ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 2: LAB INVESTIGATION: SYMMETRIC SPLITTING

CONCRETE EXPLORATION • PHYSICAL MEASUREMENTS

STACKCALC LEARNING LAB

LEVEL 1 & LEVEL 2 HANDS-ON INVESTIGATION • THE CONCRETE DISCOVERY

STEP-BY-STEP LABORATORY EXPLORATION:

Step 1: If you put all 6 x-rods on the right, you get an elongated rectangle, not a square.

Step 2: To keep the shape square, divide the 6 x-rods EQUALLY: 3 along the top and 3 along the side (b/2 = 6/2 = 3).

Step 3: Look at the bottom-right corner: There is an empty square hole of dimension 3 by 3!

Step 4: Fill the hole with unit tiles: Exactly 3 × 3 = 9 unit tiles are needed to COMPLETE the square! Result: $(x + 3)^2 = x^2 + 6x + 9$.

KEY PHYSICAL INVARIANTS TO OBSERVE & RECORD:

✓ Invariant 1: Physical dimensions match mathematical truth. No visual guessing or ruler errors.
✓ Invariant 2: Rotating, swapping, or regrouping blocks preserves total height, area, and sum.
✓ Invariant 3: The whole frame datum acts as a permanent self-checking boundary.
✓ Invariant 4: Reverse the action physically to verify inverse operations (Subtraction & Division).

CALIBRATION & DATUM TARGET

MITUTOYO ±0.05 MM CALIBRATION

Target Datum:
Square Completion: $x^2 + 6x + 9 = (x + 3)^2$

PHYSICAL APPARATUS MANIFEST:

1 x^2 flat, 6 x-rods, 9 unit tiles + SC-32 Hardware Brent Solver

💡 **LABORATORY PRO-TIP:**

Always hold blocks strictly perpendicular to the desk. The snap detents are engineered for a clean 6N tactile insertion force. If parts wedge or tilt, inspect the socket floor for stray dust or misalignment.

Level check: Eye-level sighting is accurate to ±0.2 mm.

---- INSIDE LAB SPREAD (STUDENT FACING WHEN TABLE TENT IS OPEN) ----

ALGB-01 • TENT 6/6 | ADVANCED | 25 Min

PANEL 3: LEVEL 3 CHALLENGE: SOLVING $x^2 + 6x = 16$

NON-ROUTINE INQUIRY • RPN HARDWARE STACK EXECUTION

STACKCALC LEARNING LAB

LEVEL 3 CHALLENGE: SOLVING $x^2 + 6x = 16$

THE NON-ROUTINE PUZZLE:

Use your completed square to solve the equation $x^2 + 6x = 16$.

FORM A MASTER STATION TASKS (Complete All 3):

1. **Add 9 to both sides: $(x^2 + 6x) + 9 = 16 + 9$.**
   Expected Physical Finding: $(x + 3)^2 = 25$

2. **Take square root of both sides: sqrt((x + 3)²) = sqrt(25).**
   Expected Physical Finding: $x + 3 = \pm 5$

3. **Solve for x: x = −3 + 5 and x = −3 − 5.**
   Expected Physical Finding: $x = 2$ or $x = -8$

RPN CALCULATOR STACK TRACE (SC-6 & SC-32 HARDWARE)

CANONICAL KEYSTROKES: [x^2 + 6x − 16 = 0] [SOLVE] –> x = 2, x = −8

KEY	REGISTER STATE (T, Z, Y, X)	MATHEMATICAL STACK ACTION
eqn	X: $x^2 + 6x - 16$	Define quadratic equation
SOLVE	X: 2 | Y: −8	Brent-Dekker root solver finds real roots via square completion

CONCRETE-TO-ABSTRACT BRIDGING PRINCIPLE:

- The physical manipulative is not separate from the calculator—it is an exact physical mirror.
- Stacking a block corresponds 1:1 to pressing [ENTER] and lifting a number into register Y.
- Pressing an operator key ([+], [x], [−], [/]) corresponds to leveling two blocks together.
- In standard testing, students who mastered the physical blocks recall the register stack spatially, making parentheses syntax errors completely impossible.

Mastery Benchmark: Complete all 3 Form A tasks and execute RPN trace in <15 minutes.

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